

SOC3010-OPERATING SYSTEM

TERM PROJECT IMPORTANT

GUIDELINES FOR DESIGN & IMPLEMENTATION

- **THE SELECTED PROJECT MUST BE IMPLEMENTED ON LINUX SYSTEM ONLY.**
- **ANY PROJECT DEVELOPED ON WINDOWS PLATFORM WILL BE GIVEN A GRADE 0.**
- **PLEASE NOTE THAT THIS IS NOT A WINDOWS BASED PROJECT. USE OF WINE IS TOTALLY PROHIBITED.**
- **THERE ARE A NUMBER OF GUI TOOLS AVAILABLE ON LINUX, STUDY THEM AND USE THEM TO DEVELOP GUI.**
PLEASE BE AWARE THAT CHEATING, COPYING AND FOOLING AROUND ARE NOT ALLOWED IN THIS TERM PROJECT.

- **YOU ARE REQUIRED TO CARRY OUT YOUR TERM PROJECT SINCERELY WITH MUCH SERIOUSNESS FOLLOWING STRICTLY ALL THE INSTRUCTIONS AND GUIDELINES GIVEN.**
- **ALL TEAM LEADERS SHOULD VISIT DURING OFFICE HOURS AND TAKE A PRIOR APPOINTMENT THROUGH EMAIL TO HAVE A DETAILED DISCUSSION WITH ME REGARDING THEIR TERM PROJECT SO THAT TERM PROJECT PROPOSAL CAN BE PREPARED IN A BETTER WAY BY INCLUDING ALL THE FEATURES DISCUSSED DURING THE MEETING.**
- **ONLY AFTER RECEIVING THE TERM PROJECT PROPOSAL DOCUMENT, A DECISION WILL BE TAKEN WHETHER YOUR TEAM SHOULD BE ALLOWED TO DO THE TERM PROJECT OR NOT.**

IMPORTANT NOTE:

- **AS WE HAVE ALL THE SOURCE CODE AND REPORTS OF ALL THE TERM PROJECTS CARRIED OUT BY STUDENTS FROM 2016 TO 2023, ALL SOURCE CODE AND REPORT SUBMITTED BY EVERY TEAM WILL BE THOROUGHLY CHECKED FOR PLAGIARISM.**
- **IT IS HEREBY INSTRUCTED THAT COPYING AND USING THE SAME TERM PROJECT SOURCE CODE AND REPORTS ARE TREATED AS ACTS OF CHEATING & PLAGIARISM AND GIVEN A GRADE OF 0 FOR TERM PROJECT.**
- **YOU ARE HEREBY WARNED NOT TO INDULGE IN SUCH ACTS OF CHEATING AND PLAGIARISM**

PROJECT DESIGN & IMPLEMENTATION

- **EACH PROJECT TEAM IS REQUIRED TO DEVELOP THE SOFTWARE MODULES THAT MODEL THE BUSINESS PROCESS ACCORDING TO THE SYSTEM REQUEST.**
- **PLEASE NOTE THAT THIS TERM PROJECT IS NOT A WEBSITE DEVELOPMENT PROJECT, IT IS PURELY A DESKTOP APPLICATION PROJECT. HENCE YOU SHOULD NOT USE ANY WEBSITE**

DEVELOPMENT TOOLS LIKE WEBPRESS ETC.

- **PROJECT MUST BE IMPLEMENTED USING CLIENT – SERVER ARCHITECTURE WITH SERVER ON ONE MACHINE AND CLIENTS IMPLEMENTED ON DIFFERENT MACHINES.**
- **FOR EXAMPLE, IF A TEAM HAS 8 MEMBERS INCLUDING THE LEADER, THEN SERVER MUST BE SETUP ON ONE MACHINE AND CLIENTS MUST BE SETUP ON SEVEN MEMBERS' LAPTOPS.**
- **IF SERVER AND CLIENTS ARE SETUP ON THE SAME LAPTOP THEN THE TERM PROJECT WILL BE TREATED AS A SIMPLE HOME ASSIGNMENT AND GIVEN A GRADE OF 0 FOR THE PROJECT WORK.**
- **THE PROJECT INVOLVES THE DESIGN AND DEVELOPMENT OF THE FOLLOWING THREE COMPONENTS:**
 - **FRONT END**
 - **MIDDLE LAYER**
 - **BACK END**

FRONT END COMPONENT (CLIENT SIDE):

- **THE FRONTEND GUI AT CLIENT SIDE MUST BE DEVELOPED USING THE GUI TOOLS AVAILABLE ON**

THE LINUX SYSTEM (USING GLADE, GTK+3 OR ANY OTHER TOOLS OF YOUR CHOICE ON THE LINUX SYSTEM)

MIDDLE LAYER COMPONENT (BOTH CLIENT & SERVER SIDE):

- **ALL INTER PROCESS COMMUNICATIONS BETWEEN SERVER AND CLIENTS SHOULD BE ORGANIZED USING THE IPC MECHANISM -SOCKETS (SOCKETS PROGRAMMING WITH MULTIPLE CLIENTS ON DIFFERENT MACHINES).**
- **THE INTER PROCESS COMMUNICATION BETWEEN SERVER AND CLIENTS USING SOCKETS SHOULD BE IMPLEMENTED USING C LANGUAGE.**

BACKEND COMPONENT (SERVER SIDE):

- **BACKEND DATABASE MUST BE DEVELOPED ON THE SERVER USING ANY ONE OF THE FAMILIAR DATA BASE MANAGEMENT SYSTEM TOOLS (MySQL, MariaDB, PostgreSQL, SQLite, or even DB2, ORACLE, MaxDB) AVAILABLE ON THE LINUX SYSTEM.**
- **ALL THE DATA REQUIRED BY THE CLIENTS AND GENERATED BY THE CLIENTS MUST BE STORED ON THE DATABASE TABLES AT THE SERVER SIDE.**
- **ALL QUERIES BY THE CLIENTS AT THEIR CLIENT TERMINALS MUST BE SATISFIED BY ACCESSING THE DATABASE TABLES AT THE SERVER OVER THE NETWORK.**

FOR PROCESSING AT THE SERVER SIDE, YOU CAN USE ANY PROGRAMMING LANGUAGE OF YOUR CHOICE, BUT IT DEPENDS ON THE DATABASE TOOL USED AT THE BACKEND.

- **AT THE SERVER SIDE, A CONNECTION WINDOW SHOULD BE DEVELOPED TO SHOW THE CONNECTIONS MADE TO THE SERVER BY THE CLIENTS. WHENEVER A CLIENT MAKES A REQUEST FOR A CONNECTION AND GETS CONNECTED TO THE SERVER AFTER ITS ACCEPTANCE. THIS WINDOW SHOULD SHOW THE CONNECTION MADE BY THE CLIENT WITH ITS IP ADDRESS AND PORT NUMBER**
- **TO ACCESS THE SERVER ALL CLIENTS SHOULD HAVE LOGIN NAME AND PASSWORD OR IN OTHERWORDS, THEY SHOULD BE REGISTERED THROUGH A REGISTRATION WINDOW AT THE CLIENT SIDE.**
- **EVEN A SYSTEM ADMINISTRATOR SHOULD HAVE A SYSADMIN WINDOW AND SYSTEM ADMINISTRATOR CAN ALSO LOGIN THROUGH THE CLIENT TERMINAL. HAVING A SYSADMIN WINDOW AT THE SERVER SIDE IS ALSO MOST DESIRABLE TO UPDATE THE DATABASE TABLES AND MANAGE ALL ADMINISTRATIVE TASKS.**

PROJECT DESIGN & DEVELOPMENT:

- **AS YOU HAVE SELECTED THE PROJECT TITLE, YOU ARE REQUIRED TO UNDERSTAND ALL THE FEATURES REQUIRED TO BE PROVIDED AT THE CLIENT SIDE AND SERVER SIDE, IDENTIFYING VARIOUS COMPONENTS OF THE SYSTEM WHICH YOU ARE DEVELOPING AND TASKS TO BE PERFORMED IN THESE COMPONENTS OF THE SYSTEM AT THE SERVER SIDE AND AT THE CLIENT(S) SIDE**

- **IN ORDER TO DEVELOP THESE MODULES, YOU ARE REQUIRED TO LOOK CLOSELY AT THE SYSTEM IN REAL LIFE AND THE SERVICES OFFERED BY THE REAL-LIFE SYSTEM TO THE CLIENTS AND ALL THE REQUIREMENTS OF THE CLIENTS BY CARRYING OUT DEEP DESIGN AND ANALYSIS OF ALL THE COMPONENT MODULES REQUIRED FOR BUILDING THE SYSTEM.**
- **IT IS ADVISABLE TO CREATE TERM PROJECT REQUIREMENT DEFINITIONS IN THE FORM OF USE CASE DIAGRAM, ACTIVITY DIAGRAMS, CLASS DIAGRAMS, BEHAVIORAL DIAGRAM WHEN YOU CARRY OUT THE DESIGN AND DEVELOPMENT OF VARIOUS COMPONENTS OF YOUR SYSTEM**
- **KNOWLEDGE OF C PROGRAMMING, JAVASCRIPT/JASON SCRIPT, GUI TOOLS AND DATABASE TOOLS ON LINUX IS HIGHLY DESIRABLE**

PROJECT LIVE DEMO

- **THE PROJECT LIVE DEMO MUST BE PROVIDED BY SETTING UP SERVER ON ONE LAPTOP AND SETTING UP CLIENTS ON DIFFERENT SYSTEMS (MEMBERS LAPTOPS).**
- **FOR EXAMPLE: IF YOU HAVE A TEAM OF 8 MEMBERS THEN ON ONE LAPTOP YOU MUST SETUP THE SERVER AND ON THE REMAINING 7 LAPTOPS YOU MUST HAVE THE CLIENTS CONNECTED TO THE SERVER.**
- **PROVIDE DEMO OF ALL THE FEATURES AVAILABLE AT THE SERVER SIDE AND ALL THE FEATURES AVAILABLE AT THE CLIENT SYSTEMS.**

- **EVERY CLIENT MUST CONNECT TO THE SERVER AND TRY TO ACCESS ALL THE FEATURES PROVIDED BY THE SERVER.**
- **PLEASE NOTE THAT YOU MAY BE ASKED TO MODIFY SOME OF THE FEATURES IN VARIOUS COMPONENT MODULES IN YOUR SYSTEM (MODIFYING EXISTING FEATURES OR ADDING NEW FEATURES TO YOUR SYSTEM) DURING THE LIVE DEMO**

PROJECT PRESENTATION, DEMO & REPORT

- **PROJECT PRESENTATION & DEMO WILL BE HELD IN THE 16TH WEEK FROM 23RD TO 28TH DECEMBER 2024**
- **A HARDCOPY OF THE REPORT MUST BE SUBMITTED BY THE TEAM LEADER JUST BEFORE THE PRESENTATION ON THE DAY OF THE PRESENTATION OR ONE DAY BEFORE THE PRESENTATION.**
- **PLEASE NOTE THAT WITHOUT THE TERM PROJECT REPORT, NO PRESENTATION WILL BE ALLOWED AND GIVEN A GRADE 0 FOR THE TERM PROJECT**
- **PROJECT REPORT & DETAILED PRESENTATION GUIDELINES WILL BE PROVIDED AT THE ECLASS PORTAL**

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ULTIMATE OBJECTIVE & GOAL:

- **THIS TERM PROJECT IS ASSIGNED TO YOU ALL TO**
 - **PROVIDE AN OPPORTUNITY TO WORK IN A TEAM TO ENHANCE TEAM WORKING SKILLS,**
 - **DEVELOP TEAM LEADERSHIP QUALITIES,**
 - **INCULCATE CRITICAL THINKING ABILITIES,**
 - **DEVELOP SELF-LEARNING SKILLS, AND**
 - **ABOVE ALL TO PROVIDE A RICH EXPERIENCE OF DESIGNING AND IMPLEMENTING A LIVE REAL-LIFE PROJECT IN A TEAM.**

WISH YOU ALL THE BEST